

HIMANSHU MUDIGONDA

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SUMMARY

Computer Science undergraduate (B.Tech, 2027) with strong foundations in C/C++, Java, and OOP design, and hands-on experience building, debugging, and owning full-stack software components end to end. Comfortable across the SDLC — requirements, design, implementation, testing, and production deployment — with practical exposure to distributed client-server systems, API/socket-based integrations, relational databases, and design-pattern-driven architecture. Strong analytical and troubleshooting mindset built through real production debugging and client-facing issue resolution; keen to apply this to complex, standards-compliant systems software in the semiconductor and wafer-fabrication equipment domain.

TECHNICAL SKILLS

Languages: C, C++, Java, Python, JavaScript, SQL

Core CS: Data Structures & Algorithms, Object-Oriented Programming (OOP), Design Patterns, Operating Systems fundamentals

Systems & Backend: REST APIs, Client-Server Architecture, Socket-based Communication, Microservices, Distributed Systems basics

Databases: SQL Server, PostgreSQL, MySQL, MongoDB, Redis

Debugging & QA: Unit/Integration Testing, Root-Cause Analysis, Production Issue Debugging, Automated Test Suites (pytest, Vitest)

Tools & Practices: Git, Docker, CI/CD (GitHub Actions), Linux, Agile/SDLC, Technical Documentation

Web & Frameworks: React.js, Node.js, Express.js, FastAPI, Flask, Next.js

PROJECTS

FinPilot — AI Personal Finance Copilot | Next.js 14, FastAPI, PostgreSQL/TimescaleDB, Redis

- Designed and implemented a full-stack client-server system (Next.js frontend + FastAPI backend) with a 7-table relational schema on PostgreSQL/TimescaleDB, Redis caching, and Alembic-managed migrations; 87 automated tests (pytest, Vitest, ESLint, mypy) gating CI/CD before every deploy.
- Built a fault-tolerant provider-chain architecture (Groq → Gemini → Anthropic → Ollama → local) with automatic failover and graceful feature-flagged degradation on constrained hosts — direct application of resilient distributed-system design.
- Owned complete feature components independently: budget tracking, statistical anomaly/fraud detection (Isolation Forest + graph analysis), and a time-series forecasting engine (ARIMA + LSTM ensemble).
- Hardened the system end-to-end: JWT auth with refresh-token rotation, Postgres row-level security, CSRF protection, SQL parameterization against injection, and strict CORS/HSTS policy — security-by-design across every layer.

CloakDoc — Privacy-Preserving Document Sanitizer | MERN, Node.js, MongoDB, React, Docker

- Built a full-stack application performing byte-accurate, format-faithful document editing: DOCX modified in-place at the XML level and PDFs reconstructed operator-stream-accurate, requiring careful low-level parsing and precise object-model manipulation.
- Designed the system with OOP principles and a modular rule engine so new PII-detection categories plug in without touching core logic — the same extensibility discipline expected of standards-compliant, maintainable software components.
- Implemented defensive engineering throughout: JWT auth, AES-256-GCM encryption at rest, automatic data-expiry, and 22 automated tests; shipped as a one-command Docker deployment for reproducible installs.

FastAPI Finance Dashboard | FastAPI, MongoDB, React, Recharts

- Built a production REST API (12+ endpoints, OpenAPI spec, API-key auth) and engineered a bulk CSV ingestion pipeline that indexed and aggregated 500K+ records for sub-second query performance under load.
- Debugged and resolved data-consistency and performance bottlenecks during scale-testing, applying systematic root-cause analysis rather than trial-and-error fixes.

FTP Fitness Web Application | React.js, Node.js

- Owned the complete SDLC for a full-featured platform — requirements gathering, architecture design, implementation, testing, and production deployment on Vercel.
- Structured the codebase around reusable, loosely-coupled components, applying OOP and separation-of-concerns principles to keep the system maintainable as features grew.

EXPERIENCE

Volteo Maritime — Full Stack Developer Intern

- Engineered a responsive Smartport web application with Express.js + MongoDB backend integration, delivering production-ready features across multiple concurrent modules while coordinating with cross-functional teams.
- Debugged and resolved usability and integration issues on the live APMB (Andhra Pradesh Port Management System) application, improving reliability and consistency for end users.
- Documented technical workflows and system logic into clear, structured reports for non-technical stakeholders — translating complex engineering detail into actionable communication.

Corizo — AI/ML Trainee | May 2024 – June 2024

- Completed an intensive technical training program with hands-on implementation of real-world applications, strengthening applied problem-solving and analytical fundamentals.

EDUCATION

B.Tech, Computer Science and Engineering | Amrita Vishwa Vidyapeetham, Coimbatore

2023 – 2027 | CGPA: 7.46 (74.6%)

LEADERSHIP & ACHIEVEMENTS

- **ICPC Hackathon Committee** • **Toastmasters** (communication & presentation) • **Gold Medalist** Powerlifting • **Bronze Medalist** Skating • **NCC Cadet**